



Ahmedabad
University

CSE623 : Machine Learning

**Machine Learning-based Estimation of Muscle
Cross-Sectional Area (ACSA) from Segmented
Ultrasound Images of Lower Limb Muscles for
Sports Performance Analysis**

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Abstract

The size of muscle and the capacity to generate force is a significant measure in the analysis of sports performances, and the use of anatomical cross-sectional area (ACSA). However, ultrasound imaging is normally applied in determining the morphology of the muscles, but manual estimation of ACSA using ultrasound images is tedious and operator-based. This project investigates a classical machine learning solution to predicting ACSA of segmented ultrasound images of the Rectus Femoris (RF) and Vastus Femoris (VF) muscles. Regression models that are appropriate to small datasets are applied to interpretable geometric and shape-based features obtained out of segmentation masks. The offered solution is expected to enhance objectivity and repeatability of muscle ACSA estimation using ultrasound.

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1 What We Have Done

- Researched the literature that was already published on the application of ultrasound imaging in measuring muscle morphology and anatomical cross-sectional area (ACSA) with special emphasis on lower-limb muscles, including Rectus Femoris and Vastus Lateralis.
- Considered previous studies on automated/semi-automated computational methods of muscle analysis by use of ultrasound images in order to learn about the available methodologies and constraints.
- Compared the given data in the form of normalized ultrasound images, matching segmented muscle areas, feature sets that were extracted, as well as ground-truth ACSA values.
- Studied and classified the provided features and determined their applicability as:
 - Segmentation mask induced geometric and shape features.
 - Features of intensity- and image-based that are computed in the muscle area of interest.
 - Completed the use of a feature based machine learning pipeline including the exclusion of segmentation and regression based estimation of ACSA.
 - Fixed an entire machine learning pipeline with feature validation, normalization, regression modeling, evaluation and interpretability.

2 What We Aspire to Do

- Prepare and confirm the given datasets of features of the Rectus Femoris (RF) and Vastus Lateralis (VL) muscles.
- Do preprocessing of the feature data, which may include checking of consistency, normalization and treatment of outliers where possible.
- Use classical and baseline regression methods, i.e. linear regression, regularized regression, support vector regression (SVR) and tree-based regression methods to predict ACSA.
- Measure model performance with the help of corresponding statistical measures like mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2).
- Assess the significance of features and model interpretability to interpret the role that geometric and intensity-based features make in predicting ACSA.
- Determine sensitivity of model predictions to changes in the extracted features and a segmentation derived measurements.

References

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